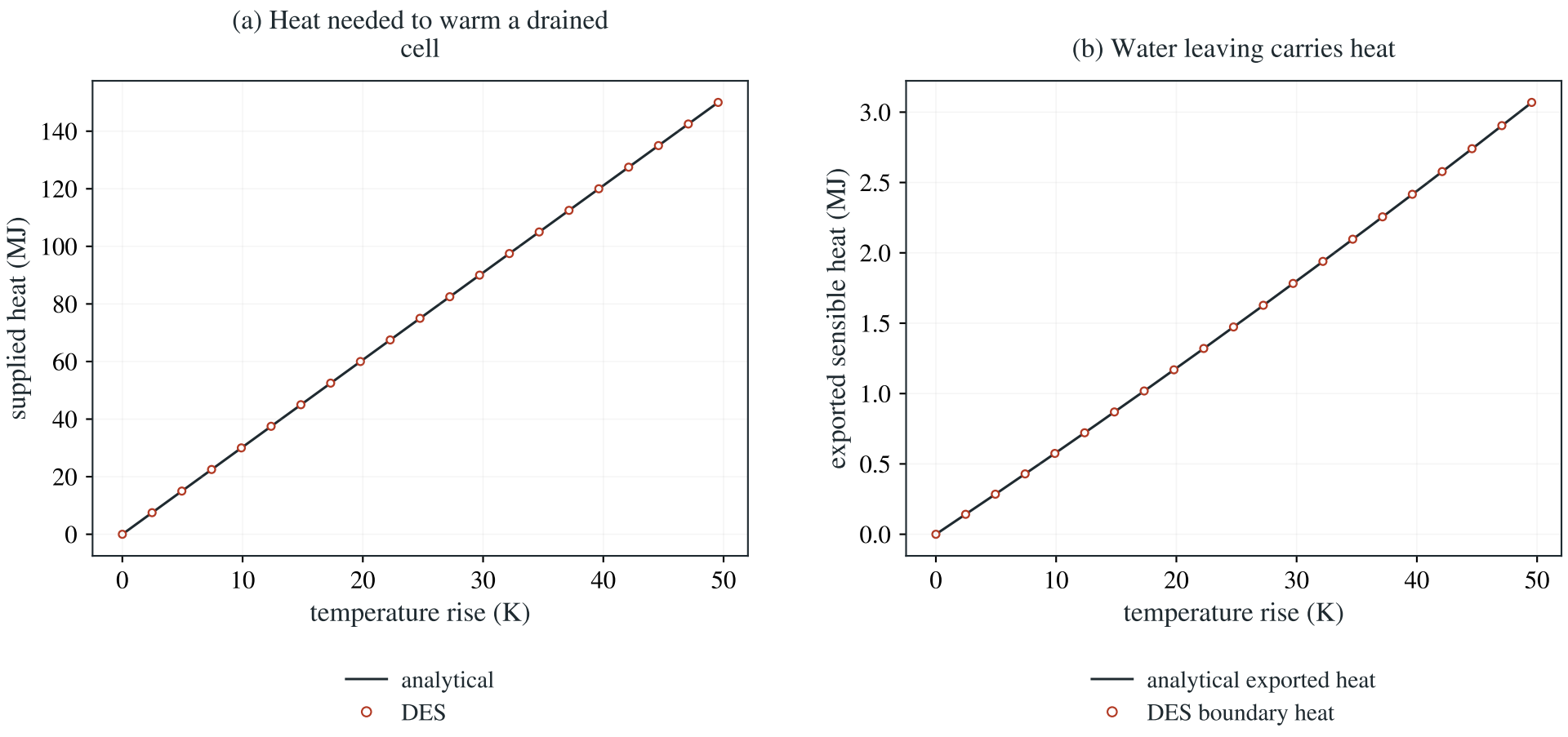

(a) Heat needed to warm a drained cell
supplied heat (MJ)
0
20
40
60
80
100
120
140
0
10
20
30
40
50
temperature rise (K)
analytical
DES
(b) Water leaving carries heat
exported sensible heat (MJ)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
10
20
30
40
50
temperature rise (K)
analytical exported heat
DES boundary heat